

The Utility Problem

By design, the electricity distribution grids are built as a 3-phase system. In some utilities, many of the consumption points are connected to only one phase. This is achieved by allocating consumption points with either phases A, B, or C of the final 3-phase power distributed for the end consumption. A present problem confronting electrical power distribution is alignment between phase identification in utility records and in reality in the field.

Historically, the phase identification information data were recorded and kept in a system of records - usually Geographical Information System (GIS). It is not different now; however, as with any data system, this system might diverge with time if corrective processes are not in place. Utilities have started to notice data errors, and one of the target areas is the correction and improvement of phase identification data.

The data errors in phase identification records might bring multiple problems. Without proper phase identification, analysts might not be able to calculate phase load correctly. Unbalanced phases lead to significant energy losses and sharply reduced asset lifetimes (due to the overloading of one or two phases). Errors in phase identification records can also lead to outages and safety issues.

Above are some of the reasons why it is so important to have a tool and process that checks that the phases are correctly identified (according to the actual field wiring).

The Data Solutions and Methods

For this use case, there are multiple data solutions and methods which can be utilized to address this issue. The following section will describe some of the possible options.

Method 1.: Voltage signature clustering:

Phase identification by voltage profiles also has multiple methods (options). The method described uses the Spectral Clustering approach. The core concept behind this approach is that the voltage profiles from AMI meters can be clustered by phase using a measure of correlation or affinity between pairs of historical time-series measurements. Pairs of voltage time-series that are on the same phase will tend to have more similar variations in voltage measurements than two time-series that come from customers on separate phases.

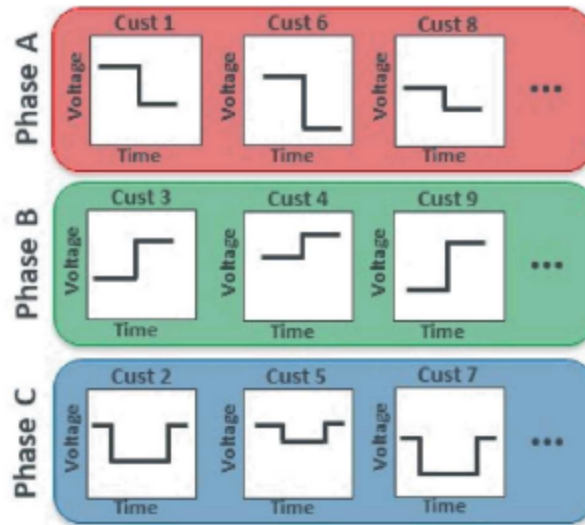


Figure 1 - Voltage correlation examples

Required data for this method are:

- GIS data connectivity information which allows for geospatial connectivity tracing:
 - Using GIS tracing, it is required to locate all the meters on the same feeder and same phase.
- Time-series data of electricity meters with voltage registers. AVG, MIN, and MAX voltage values might be beneficial as well:
 - The time synchronization of the measured data is also very important as the data of each time stamp is compared in this approach.

Method 2.: Phase Identification Inheritance:

This method is based on the fact that the MV/LV transformers are often single-phase (for the North American grid). If the meter has a different phase compared to the parent transformer and other meters under the same transformer, then most likely, the meter has the wrong phase identification.

Required data for the above method:

- Transformer and meters phase data
- GIS data connectivity information which allows for geospatial connectivity tracing:
 - To trace downstream from the examined transformer.

Phase Identification Analysis

[UC36-01] - Awesense Energy Transition Platform Use Case

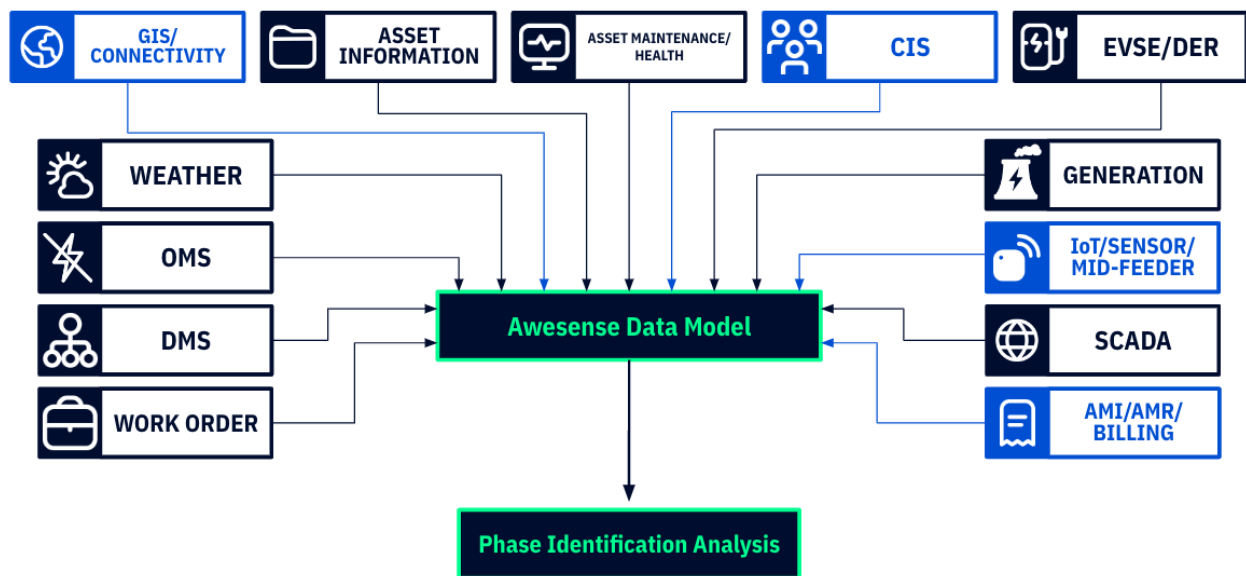


Another method, or rather the indication of errors in phase identification data on the feeder level, can be comparing Connected kW Loads vs. Actual Phase Balance (read more in Awesense Use Cases Library). With this method, a high-level overview of the potential errors can be obtained and feeders with the highest discrepancies can be targeted with methods from this use case.

Awesense Energy Transition Platform has all the data required for this use case, plus it comes with easy access techniques which allow any analyst simple and reliable access to the required datasets. With that, the actual implementation of the use case is painless as possible.

Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.



Available Use Case Implementations:

Users of the Awesense Energy Transition Platform can access this use case by the following means:

1. A GitHub link to the Jupyter notebook can be provided upon request.